

Evan Williams

☎ +1 (571) 377-9544 | ✉ evanw0405@gmail.com | 💻 [evanwilliams02](https://github.com/evanwilliams02) | 🌐 [evanmwilliams](https://github.com/evanmwilliams) | 🌐 [evanmwilliams.github.io](https://github.com/evanmwilliams.github.io)

EDUCATION

- Cornell University, College of Engineering** Ithaca, NY
Master of Engineering in Computer Science January 2024 – May 2024
- **Advisor:** Adrian Sampson | **Project:** Compiling PyTorch Models to Hardware Accelerators
- Cornell University, College of Engineering** Ithaca, NY
Bachelor of Science in Computer Science, Electrical and Computer Engineering August 2021 – December 2023
- **Cumulative GPA:** 3.72
- Purdue University, College of Engineering** West Lafayette, IN
Bachelor of Science in Computer Engineering – Transferred August 2020 – August 2021
- **Activities and Societies:** Purdue Duality Lab, Purdue Engineering Student Council, ECE Ambassadors

RESEARCH EXPERIENCE

- Stanford University** December 2025 - Present
Collaborating Researcher, Advised by Olivia Hsu (CMU) Stanford, CA
- Early-stage exploration of programming by navigation for interactive schedules in FuseFlow compiler for PyTorch
- Cornell Computer Systems Lab** January 2022 – August 2024
Undergraduate Research Assistant, Capra Research Group Ithaca, NY
- Designed compiler for a feed-forward neural network in PyTorch directly to FPGA-executable Verilog design
 - Added support to Calyx for floating-point arithmetic and wrote optimization passes for MLIR and CIRCT IRs
 - Built a pangenomic graph query accelerator for node-depth algorithm and presented findings to research group
- Purdue Duality Lab** September 2020 – May 2021
Undergraduate Research Assistant West Lafayette, IN
- Analyzed domain-specific regular expressions for DFA complexity, portability, and presence of back references
 - Built prototype search-engine over millions of regexes to enable identification and reuse of secure patterns
 - Implemented ranking scheme to cluster groups of regexes using metrics such as F-Score and mistake rate

PROFESSIONAL AND LEADERSHIP EXPERIENCE

- Amazon Web Services** August 2024 – Present
Software Development Engineer, Amazon HealthOmics Mountain View, CA
- Delivered lazy loading to download outputs from call cache for WDL engine yielding 11% speedup on workflows
 - Expanded support for CUDA driver integration and implemented type coercion to enable more customer use cases
 - Led knowledge sessions teaching engineers about Rust and wrote onboarding materials for new team members
- Cornell Bowers College of Computing and Information Science** May 2023 – August 2024
Course Developer Ithaca, NY
- Created assignments, lesson materials, and infrastructure for re-design of Cornell's core systems course, CS 3410
 - Implemented optimized MNIST classifier in C and profiled performance using Intel TopDown for new project
 - Designed assignments and created tooling for graduate computer architecture and systems course, ECE 5755
- Cornell Data Science Project Team** February 2022 – May 2024
President Ithaca, NY
- Directed team of 70+ members in ideation and implementation of machine-learning applications and research
 - Secured over 15k in corporate sponsorships and launched partnership projects with companies in industry
 - Led implementation of facial recognition system with distributed inference and federated learning on edge devices
- Amazon Web Services** May 2023 – August 2023
Software Development Engineer Intern, AWS IoT RoboRunner Sunnyvale, CA
- Developed application to perform deep packet inspection on MQTT network traffic using C++ and Docker
 - Integrated with Amazon CloudWatch and AWS DeviceDefender to alert customers of anomalous behavior
 - Deployed to AWS SnowDrop container for health monitoring and forwarding of detected anomalies to S3 bucket
- Deloitte Consulting LLP** June 2022 – August 2022
Solutions Engineering Summer Scholar New York, NY
- Produced designs and test plans for mobile app to track power outages and implemented backend and REST APIs

PUBLICATIONS

Jiahao Xie, **Evan Williams**, Adrian Sampson. "From PyTorch to Calyx: An Open-Source Compiler Toolchain for ML Accelerators". (In Submission) *Compilers for Machine Learning (C4ML) Workshop at the International Symposium on Code Generation and Optimization (CGO) 2026*.

TEACHING EXPERIENCE

CS 3110: Data Structures and Functional Programming <i>Cornell University</i>	Spring 2022, Spring 2024 <i>Teaching Assistant</i>
ECE 4750/CS 4420: Computer Architecture <i>Cornell University</i>	Fall 2023 <i>Teaching Assistant</i>
CS 4820: Intro to Analysis of Algorithms <i>Cornell University</i>	Fall 2022, Spring 2023 <i>Teaching Assistant</i>
CS 159: C Programming <i>Purdue University</i>	Spring 2021 <i>Teaching Assistant</i>

HONORS, AWARDS, AND FELLOWSHIPS

- NSF CSGrad4US Fellowship (August 2025): **\$159,000** over 3 years to support PhD studies
- Cum Laude, Cornell University (December 2023)
- Outstanding Undergraduate Course Staff, Cornell CIS (May 2022 - December 2023)
- Dean's List, Cornell Engineering (August 2021 - December 2023)
- Undergrad Architecture Mentoring Workshop (uArch) Attendee at ISCA 2022 (June 2022)
- Victor H. and Helen T. Green ECE Scholarship, Purdue ECE (May 2021)